JAIDEV EDUCATION SOCIETY'S

J D COLLEGE OF ENGINEERING AND MANAGEMENT

KATOL ROAD, NAGPUR

Affiliated to DBATU, RTMNU & MSBTE, Mumbai

Website: [illegible] E mail: [illegible]

Education to Eternity

An Autonomous Institute, with NAAC "A" Grade

Department of Computer Science & Engineering

"A place to learn; A chance to grow"

Session: 2024-25

MID SEM -2 EXAMINATION

Course: B. Tech. in Computer Science & Engineering **Year/Semester:** 2nd /4th

Subject: Formal Languages & Automata Theory **Subject Code:** CS4T001

Max Marks:20 **Date: 09/04/2025** **Duration: - 1 Hr.**

Instructions to the Students:

1. **All the questions carry marks as indicate.**
2. **Assume suitable data whenever necessary.co1**
3. **Illustrate your answer whenever necessary with the help of neat sketches.**

Students should be able to:

CO1. Define the mathematical principles behind theoretical computer science.

CO2. Differentiate and give examples for the different types of automata like finite automata, push down automata, linear bounded automata and Turing machine.

CO3. Correlate the different types of automata to real world applications.

CO4. Choose and design appropriate automata for the different requirements outlined by theoretical computer science.

CO5. Identify the different computational problems and their associated complexity.

SECTION A

1. **All Questions are compulsory.**
2. **Answer in one word/sentence.**

Q. No.	Questions	Level/CO Mapped	Marks
Q.1	Describe the language generated by the grammar G=(V,T,P,S) Where P is $S \rightarrow aSb$, $S \rightarrow ab$	L2/CO2	1
Q.2	Draw the block diagram of PDA.	L1/CO1	1
Q.3	Write the steps to reduce the CFG.	L1/CO1	1
Q.4	State the definition of Ambiguous grammar.	L1/CO1	1

SECTION B

1. **All Questions are compulsory.**
2. **Solve answer in brief (two, three Sentence)**

Q.5	Prove that the language L={ ww^R | w ∈ {a,b}8} is not regular.	L3/CO3	2
Q.6	Construct a regular grammar for RE a*b(a+b)*.	L4/CO3	2
Q.7	Find left linear grammar for (0+1)*00(0+1)*.	L2/CO2	2
Q.8	Convert the grammar in Chomsky Normal Form. S → AB | aB A → aab | ε B → bbA	L2/CO2	2

SECTION C

1. **Solve any two questions.**

Q.9	Consider the following grammar. S → 0S | 1A | 1 A → 0A | 1A | 0 | 1 Find equivalent finite automata.	L1/CO3	4
Q.10	Consider the CFG. S → aB | bA A → a | aS | bAA B → b | bS | aBB For the string 'bbaaabbaba', find i) Leftmost derivation ii) Rightmost Derivation	L2/CO2	4
Q.11	Construct a pushdown automata to accept the following language L= { $a^n b^n$ | n>=1}.	L4/CO4	4